Rescue Studio user guide

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Use this guide to turn footage into corrected labels, evaluate saved recognition runs, check your dataset, train new weights and compare models. Everything here uses the desktop buttons; you do not need terminal commands or camera calibration.

The important order: generate or draw boxes, correct them, mark the frames reviewed, then evaluate or export. Saving labels does not train the model. Only **Start training...** updates weights.

Choose your task

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Annotation toolbar	Play footage, list boxes, cut clips and review frames in batches.	4
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Click a task in this table or use the PDF bookmarks to jump to it. For evaluation and dataset checks, start with pages 5 to 8.

Open footage and get draft boxes

Resume your work

1. Open **Rescue Studio**. Choose your existing workspace when asked. Your current project workspace is **C:/Users/Hayden/Documents/clack/autonomous_tracker**.
2. Select a video in Home and click **Continue labeling**. For a new video, use **Import video...**, then **Start labeling**. Resume does not run analysis or training.
3. If your project is not listed, use **Resume project...** and select its annotation JSON. For file locations, select the footage in Home and open **Files and credits**.

Generate model predictions

1. Check the model shown in Home and its **Predicts** label list. A class you add to annotations is not automatically a class the current model can recognize.
2. Click **Generate draft boxes**, read the enabled labels and confirm. Wait for completion. This creates a saved observation run without changing your annotations.
3. Click **Refresh library** if needed. Select the desired recognition result under **Optional: import recognition boxes**, then click **Label with selected predictions...**.
4. Inside the editor, the equivalent action is **Import recognition boxes...**. Choose the run folder containing **summary.json** and **frames.jsonl**, not the video file.

Existing labels are protected during import. **Use predictions on this frame** explicitly replaces that frame's labels after confirmation. **Show original predictions** displays dashed reference boxes; these are separate from the editable labels.

Process several videos

In Home, open **Batch video processing**. Add or remove footage so the list contains exactly the videos you want. Click **Create batch from footage...**, then **Start / resume / retry unfinished**. Creating a batch alone does not run inference. It pins the current model.

Use **Review selected result** to open its draft predictions, or **Open selected attempt folder** to find the run. **Stop batch** requests a stop; resuming keeps valid completed results and retries unfinished videos. A batch does not train or approve labels.

Use the original video. Do not annotate a rendered copy with model boxes burned into its pixels. Runs and labels must refer to the same exact video file content.

Correct labels and carry them forward

Make a frame ready for training and scoring

1. Open **01 Annotate footage**. Check the whole image, including areas where the model drew nothing. Your tags define the full class scope you must label.

2. Click a box to select it. Drag its interior to move it or a corner to resize it. Use **Delete selected box** for false detections. Draw missing objects yourself; enable **Draw new box** when drawing inside an existing box.

3. For a selected box, click a tag or type its name and press Enter / **Create / apply tag**. Use **New tag...** to create a class. Deselect before choosing the tag for the next box.

4. After checking every visible object in scope, click **Mark frame reviewed**. A reviewed frame with no boxes means you checked it and none of those classes are present.

5. Click **Save project** or Ctrl+S. Use **Undo label edit** or Ctrl+Z to undo mistakes. Editing a reviewed frame returns it to draft; review it again when finished.

Changing class scope affects existing review. Adding, renaming or removing a project class returns frames to draft so you can check for the changed scope. It preserves boxes. A tag is a training class, not an extra note attached alongside another class.

Avoid redrawing every frame

Correct the current frame, choose **Follow motion**, and click **Track & next**. The editor carries the current boxes into the next frame as editable drafts. Enable **Auto-track on Next** to do this on ordinary Next clicks. It does not apply to Play or jumps.

Use **Keep positions** or **Copy all boxes to next** for a fixed-position copy. These do not check whether the objects moved or are still present. **Track range...** previews a longer range; **Update following...** propagates a correction to later untouched generated drafts while protecting manual corrections and reviewed frames.

Needs attention means the match needs checking. A red dashed box may be the last known position, not a successful track. Move/resize it, use **Confirm / update selected box** if correct, or delete it if absent. Changing its tag alone does not clear the warning.

Keep an optional **Object ID** consistent for the same object across frames. The automatic T- display ID is internal carry history, not a human truth Object ID for measuring ID changes. Tracking does not discover new objects, so add new arrivals manually.

Playback clips and batch review

Watch the footage and find existing boxes

The top annotation toolbar has **Play / Pause**, a speed selector, a seek slider, timestamp and frame number. Playback shows existing boxes but never carries them or marks frames reviewed. Click the image to pause and edit. Playback has no audio; slow decoding can reduce the actual playback speed.

All boxes... lists box occurrences throughout the current project, including unsaved edits. Search by tag, ID or frame and filter Draft, Reviewed or Needs attention. Double-click a row to select that box in the editor. Use **Search / refresh** after edits. Repeated IDs are occurrences across frames, not extra unique objects.

Cut a section into its own video and label project

1. Click **Cut clip...**. Navigate the main video to your first wanted frame and press **Set from current** beside In. Do the same for Out, or type the frame numbers.
2. Choose a destination and click **Export clip + labels**. Both In and Out are included. Frame numbering begins at zero in the exported clip.
3. Wait for verification, then use **Open exported project**. The new folder contains clip.avi, clip.json and clip-origin.json. Originals are not shortened or overwritten.

The clip keeps boxes, IDs, warnings and review states from the snapshot at export start. Later edits to the original do not update it. Export is lossless video only, without audio, so files can be large. Keep the clip in the same dataset split as its source.

Review several frames at once

Click **Bulk review...**, set the range, and **Show / refresh range**. Inspect the six thumbnails; click one to correct its frame in the main editor, then refresh. Use **Mark this page reviewed...**, **Mark entire range reviewed...** or **Return range to draft...**. Read the confirmation counts before approving.

Entire-range approval can include frames not on the current page. Empty/unlabeled frames are excluded unless you explicitly include reviewed negatives. Unresolved Needs attention frames stay blocked. One Undo restores a batch; Save persists it.

Use a suggested review queue

In **05 Assisted review**, choose a recognition run or **Without predictions**, then **Build suggested queue**. Use **Open next suggested** and correct all scoped objects. **Review and next** saves and advances within the queue. Skip does not approve a frame. Finishing the queue does not mean the whole video is reviewed.

Evaluate a saved run

Use **02 Evaluate runs** when you already have saved model predictions and want to know how they compare with your corrected labels. You can evaluate a baseline before training, then evaluate new predictions after training.

Before you click evaluate

You need the original video, its annotation project with at least one reviewed frame, and a saved run folder containing **summary.json** and **frames.jsonl**. If you have no run yet, return to Home and use **Generate draft boxes** as described on page 2.

Correct the labels first. Imported predictions must not simply be marked reviewed without inspection: otherwise you can end up scoring the model against its own mistakes.

Click by click

1. In **01 Annotate footage**, check your chosen frames, mark them reviewed and save. Include frames with difficult objects and genuine negative frames, not just easy detections.

2. Open **02 Evaluate runs**. Enter the **Incident / recording group** and choose the **Dataset split**. Use the same incident name for related recordings.

3. Click **Add saved run and evaluate**. Select the run's folder, not an MP4, annotation JSON, dataset folder or training checkpoint.

4. Wait for a new row in the run table. Select it to see **Metrics and provenance**, including per-class scores, coverage, timing notes and the saved result location.

5. Check **Coverage** before judging the scores. Then open **Inspect error frames** and look at the actual misses and false detections. Pages 6 and 7 explain these results.

6. To compare another model or configuration, generate its predictions on the same video and add that saved run here too. Keep the same reviewed labels for both.

What changes and what stays unchanged

Evaluation saves the open project and creates a new result snapshot. It reads the predictions already on disk; it does not rerun YOLO, train weights or replace your boxes. The current GUI matches same-class boxes at an intersection-over-union threshold of 0.5.

Results are saved in **evaluations/evaluation-...** beside the annotation project. The footer and details panel show the exact location. Failed or unfinished runs cannot be scored; an accepted stopped or frame-limited run may still have incomplete coverage.

If you correct reviewed labels later: older rows may say STALE. Add the same saved run again to rescore it against the new reviewed labels. A new inference run is needed only when you want new predictions, such as after changing the model.

Understand the scores

Metric	Meaning and how to read it
TP	True positives: same-class predictions that matched reviewed boxes.
FP	False positives: predictions without a matching reviewed box. Extra duplicate boxes can count here.
FN	False negatives: reviewed objects without a matching prediction. These are misses.
Precision	TP divided by TP + FP. Of the detections made, how many matched the labels?
Recall	TP divided by TP + FN. Of the labeled objects present, how many were found?
Coverage	Scored reviewed frames / all reviewed frames. 95/100 means five reviewed frames were not scored.
ID changes	Changes in model tracking ID for the same human Object ID across matched reviewed observations, including gaps.
p95 ms	About 95% of recorded processing times were at or below this value. Includes read, inference, rendering and encoding, not full sensor latency.
N/A	The measure is undefined or lacks the required evidence. It does not mean perfect performance.

A small worked example

Suppose there are 20 reviewed fire boxes. The model matches 15, misses 5 and adds 3 false boxes. Then **TP = 15, FN = 5, FP = 3**. Precision is 15 / 18 = **83.3%**; recall is 15 / 20 = **75%**. High precision would not erase those five misses.

Avoid misleading conclusions

IoU measures the overlap area divided by the combined area of two boxes. This app uses fixed-threshold box matching, **not mAP**. A bad box position can therefore cause both an unmatched prediction and a missed label even when the object was roughly located.

A missing prediction frame is a **coverage gap**, not automatically a model miss. A saved frame with zero detections is different: labeled objects on that scored frame count as misses. Look at per-class results; an overall average can hide failure on a rare class.

Detector confidence is not a calibrated probability of correctness. Repeated boxes across adjacent frames are not independent events, and FP boxes per labeled minute is not a count of independent alarms. A small or repetitive clip cannot establish field performance.

Inspect errors and compare runs

Find out why a score is low

1. Select a run row in **02 Evaluate runs**, then open **Inspect error frames**.

2. Keep **Errors and missing predictions only** checked to focus on problems, or uncheck it to include other frames. Double-click a row or click **Inspect selected frame**.

3. The left pane is **Human labels / scored snapshot**. The right pane is **Model detections / scored snapshot**. Green boxes matched; red human boxes were missed; pink model boxes were unmatched. Missing predictions are explicitly unscored.

4. Check whether the model is wrong, the human annotation is wrong, or both. Use **Edit this frame's labels** only to correct the actual labels, not to make them agree with an incorrect prediction.

5. Review the corrected frame again, save, and rescore every run you want to compare. Existing result snapshots keep their original labels and scores.

Compare models fairly

For a useful comparison, keep the exact source video, reviewed label snapshot, IoU threshold and scored frame set the same. The details panel reports differences from the first run as **NOT COMPARABLE**. Resolve them before ranking models.

Changing a model can also change its supported tags or prediction settings. Check those details as well. A model that never supported smoke is not an equal smoke detector simply because both runs appear in the same table. Predictions outside your declared label scope are excluded, so keep that scope consistent.

Use separate recordings for separate jobs

Split	Use
Training	Examples the model learns from.
Validation	Held-out incidents used to compare versions and choose settings.
Test	Separate incidents reserved for a final evaluation after model selection.

Keep related recordings and derived clips together. For example, incident A can be training, incident B validation, and incident C test. Different filenames or chunks of one recording are not independent examples.

Next: inspect misses and false detections for each important class. Correct annotation errors, label examples of the failure conditions, then train and compare a new version.

For a final test, generate predictions with the selected model on reserved test footage and use Evaluate runs. Do not repeatedly tune against this test set; the guided comparison on page 9 is for validation.

Check and organize the dataset

1. Save every project you intend to check, including projects open in other windows. The check saves the current editor, then reads the other projects from disk.
2. Open **03 Dataset checks** and click **Select annotation projects**. Select the current annotation JSON files together. This replaces the selection, so include the current project too if needed. Do not choose backups, truth exports or run summaries.
3. Click **Check dataset splits**. Read the report, issue codes and affected project paths. The saved report location appears in the status line.
4. To fix assignments, open **04 Export and train**, add the relevant projects, select each row, enter its incident and split, then click **Save selected assignment**. Return to Dataset checks and run it again.

Finding	What to do
Source or incident crosses splits	Keep all recordings from the same source/incident in one split. Do not rename an incident to hide overlap.
Clip crosses splits	Move the clip assignment to the same split as its parent and related clips.
Duplicate source or related clips	Remove duplicate revisions from the selected dataset. Avoid double-counting overlapping clips.
Missing incident or unassigned split	Assign a meaningful recording group and training, validation or test.
Different class scopes	Use Edit shared dataset tags... in Export and train, then recheck frames for the full scope. Changed scopes return frames to draft.
No reviewed frames	Correct frames, mark them reviewed and save before exporting.
No reviewed negatives	Look for genuine frames with none of the scoped classes. Review those explicitly; never erase real objects to create negatives.
Invalid project	Select a real annotation project, not a report or export. Preserve a damaged file and recover a valid backup if needed.

CONFLICTS_FOUND means an error was found. **NEEDS_REVIEW** means there are warnings. **NO_KNOWN_OVERLAP** means only that these checks found no issue. A warning can still block training export when it violates an export requirement.

This auditor checks saved hashes, recorded clip ancestry and normalized incident tags. It does not open the footage to inspect labels or prove data independence or rights. Training export separately verifies the selected source videos. Reports are saved under **audits** beside the open annotation project.

Export train and test a model

Create the training dataset

1. Open **04 Export and train**. Click **Add annotation projects...** and choose one current project per source video. Assign incidents and splits, then save each assignment.

2. Check that all projects use the same full tag scope. You need reviewed training and validation projects from separate incidents, and at least one reviewed training box for every class. These are minimum software requirements, not enough data for reliable performance.

3. Click **Check and export...** and choose a destination. The app creates a new dataset with reviewed images and YOLO labels. Reviewed negative frames get empty label files. Draft frames are excluded. Wait for a successful export before continuing.

An export is a snapshot. If you change reviewed labels, scope, split assignments or the project selection, **STALE EXPORT** means you must export again. Do not edit the generated dataset files manually. **Export reviewed JSON** in the annotation tab is a different ground-truth export; it is not the training dataset action.

Train new weights

1. Click **Choose trusted .pt weights...** and select a local detection checkpoint. Only load files from a trusted source. The existing starter workspace has **models/rescue-fire-smoke/best.pt**; an empty installation does not include weights.

2. Review epochs, image size, batch and device. The installed build uses **cpu**. An epoch is a pass through training examples; batch is how many images are processed together. Image size must be a multiple of 32. More epochs do not guarantee a better model.

3. Click **Start training...** and confirm. Watch the logs and wait for completion. Use **Stop training** if needed. Training writes a separate run and does not activate the new model automatically.

Test the result on validation footage

Use **Test models and view saved comparisons** to reach step 4. After training, select a reviewed **Validation project** and click **Test this model** or **Compare with starting model**. Unlike Evaluate runs, these buttons run real inference and then score it.

The guided check needs a completed Studio training result and validation footage with matching tags. It rejects known training/final-test source or incident overlap. Always run Dataset checks across all related projects too. Unsupported starting-model tags are listed; differing class support is not a like-for-like comparison.

Use **Inspect selected run / correct boxes** to review mistakes and **Open selected result folder** for the evidence. After correcting reviewed labels, rerun the check. **Open completed training...** reopens an earlier training result.json.

Model versions files and troubleshooting

Choose which model new jobs use

After testing, use **Save this trained version to model library...**, or open **Model library and versions** from Home. Importing a model or selecting its row does not activate it. Click **Use for new observation jobs...** explicitly. **Revert to previous active version...** restores the previous active selection.

Model files are referenced, not copied. Keep the trusted .pt file and its training folder. Previously created batches keep their pinned model; create a new batch to use a new version. For a model without a Studio training record, generate predictions and use Evaluate runs.

Where your files are

Video and labels: Home > Files and credits > Open video folder / Open annotation folder. Resume project opens projects outside the library.

Prediction runs: workspace diagnostics/rescue/review-... folders contain summary.json and frames.jsonl. **Evaluations and audits:** evaluations and audits folders beside the annotation project; the editor shows the exact paths.

Cut clips: by default, clips beside the project. **Training:** your chosen dataset export folder, with training-runs inside it. Keep data.yaml, manifest.json and completed training result folders.

Common problems

Problem	Next action
No boxes appear	Check that a run exists and has been imported. Check the model's supported tags. Add missed objects manually.
Cannot evaluate	Review at least one frame; choose the run folder with both required files. Use predictions from the exact same video. Regenerate legacy runs missing a video hash.
STALE or NOT COMPARABLE	Rescore all compared runs on the same reviewed labels. Check source, frame coverage, IoU and supported classes.
Training export is blocked	Fix class-scope, reviewed-frame, incident or split problems. Remove duplicate revisions, then Check and export again.
Controls seem missing	Scroll the sidebar or training panel and enlarge the window. The playback and clip controls require version 0.5.

Before closing: save edits, finish or stop background jobs, and back up the workspace plus any externally referenced videos and weights. Keep .studio sidecars for workflow state. The app is for local annotation and observation; these scores do not certify field readiness.